

MANDALA: An E(3)-Equivariant Graph Neural Network Framework for Learning Sparse Electronic-Structure Hamiltonian and Density Matrices with Energy and Force Supervision

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Abstract

Electronic-structure calculations based on Kohn-Sham density functional theory remain indispensable in computational materials science and chemistry. Their computational cost, however, limits accessible system sizes and simulation times. It also makes systematic exploration of model architectures and training strategies on top of such data expensive. At the same time, conventional machine-learning interatomic models usually target only energies and forces, which leaves out the operator-level electronic information required for many scientifically relevant analyses. **Mandala** was developed to address this gap by providing a framework for learning sparse electronic-structure operators directly, while preserving access to derived observables and Hamiltonian-based analysis. We present **Mandala**, a modular software framework for learning block-sparse electronic-structure matrices with E(3)-equivariant graph neural networks. The framework is built around a unified representation of atom-resolved Hamiltonian, overlap, and density matrices, together with reusable abstractions for basis conversion, sparse block handling, irrep mapping, graph construction, model definition, and training. This design allows **Mandala** to support multiple electronic-structure backends, heterogeneous chemical compositions, and a wide range of neural architecture variants within one workflow.

A distinguishing feature of **Mandala** is that it does not stop at matrix

reconstruction. Predicted operators can be contracted to obtain energy observables and electron counts, differentiated with respect to atomic coordinates to obtain forces, and differentiated with respect to the simulation cell to obtain stress tensors. The predicted Hamiltonian also preserves access to Hamiltonian-based post-processing, including band-structure analysis. This allows the software to bridge electronic-structure learning and observable-guided atomistic modeling while retaining a representation tied to quantum-mechanical operators rather than only scalar targets.

Mandala combines a modular software architecture with an operator-centered learning formulation for sparse electronic-structure prediction. We emphasize the sparse matrix data model, the observable-guidance loss design, and the broad hyperparameter and architecture search space available in the framework. **Mandala** is intended to enable machine-learning workflows that go beyond conventional atomistic interatomic potentials by resolving electronic structure together with atomistic response, thereby opening a route toward modeling electronically complex condensed-matter and molecular systems within one scalable framework.

Keywords: Electronic structure, Density functional theory, Equivariant graph neural networks, Sparse matrix learning, Energy and force prediction, Scientific machine learning

PROGRAM SUMMARY

Program Title: **Mandala**

CPC Library link to program files: [TODO: to be added by the Technical Editor]

Developer's repository link: <https://github.com/Mandala-org/Mandala>

Licensing provisions: Apache-2.0

Programming language: Python

Nature of problem: Electronic-structure calculations based on Kohn-Sham density functional theory are the central computational method in computational materials science and chemistry, but their computational cost limits accessible system sizes, trajectory lengths, and large-scale model-selection studies. Atomistic simulations based on machine-learned interatomic potentials alleviate this cost for energies and forces, yet they generally do not expose the electronic operators needed for richer observables or electronic post-processing and cannot be used to model processes that involve electronic charge transfer reliably. A complementary route is to learn sparse representations of electronic-structure matrices themselves. Such a computational method must support multiple atom types, changing orbital bases,

periodic boundary conditions, equivariant geometric processing, and physically meaningful observable targets such as energies, forces, and stress.

Solution method: **Mandala** represents Hamiltonian, overlap, and density matrices as atom-pair-resolved sparse block tensors and maps these blocks to irreducible $E(3)$ representations compatible with equivariant neural networks. The package combines parser modules for electronic-structure data, basis-conversion utilities, a shared irrep mapper, graph construction routines, and configurable $E(3)$ -equivariant message-passing models. Multi-head prediction allows simultaneous learning of several matrices, while observable-level quantities are evaluated from the predicted operators using sparse traces. Observable-guidance losses can be added during training, while forces and stress are obtained through automatic differentiation of predicted energies with respect to atomic coordinates and cell vectors. The framework is organized to support extensibility, architecture search, and reproducible training workflows.

Additional comments including restrictions and unusual features: The framework is optimized for sparse matrix learning the electronic structure in atomistic systems with localized orbital bases and periodic or non-periodic geometries. A distinctive feature of **Mandala** is the combination of operator-level learning with optional observable guidance derived from those operators, which enables a single software stack to serve both electronic-structure emulation and observable-guided atomistic applications. The package is designed to accommodate multiple data backends and a wide search space of equivariant architectures and training objectives.

1. Introduction

Kohn-Sham density functional theory (DFT) is the standard method of electronic-structure simulation because it offers a practical compromise between physical fidelity and computational cost across chemistry, condensed-matter physics, and materials science [1, 2]. Even so, repeated self-consistent solution of the Kohn-Sham equations remains a major bottleneck in workflows that require large supercells, long molecular-dynamics trajectories, broad compositional screening, or repeated re-evaluation under changing thermodynamic conditions. This cost has motivated sustained interest in machine-learned surrogates for atomistic and quantum-mechanical simulation. Relevant directions include learned potential-energy surfaces and interatomic forces, as well as learned electron densities, wave-function-related quantities, kinetic-energy functionals, and reduced electronic representations [3–12].

Many successful atomistic machine-learning approaches bypass the explicit electronic structure and instead learn scalar or vector observables such as energies and forces. This strategy is highly effective for interatomic potentials, from early neural-network potential formulations to modern equivariant graph models, and recent work has shown that equivariant neural architectures are especially well suited for energy and force prediction because they can encode rotational structure directly in the model rather than relying entirely on handcrafted invariants or data augmentation [13, 14]. However, learning only energies and forces also limits direct access to the electronic structure itself, including operator-valued quantities and Hamiltonian-derived analyses such as band structure. For applications that remain fundamentally electronic-structure-centered, it is attractive to learn the operators themselves. Examples include systems in which charge transfer, orbital hybridization, defect-localized states, response to external fields, or band-structure-level analysis are central to the scientific question. In such settings, access to the underlying electronic operators is not merely auxiliary metadata but part of the modeling target itself.

Mandala is designed around this operator-learning viewpoint. The framework targets block-sparse matrix quantities arising in localized-orbital electronic-structure calculations, in particular the Hamiltonian, overlap, and density matrices. Instead of treating these objects as dense global arrays, **Mandala** represents them as atom-pair-resolved sparse blocks and combines this representation with E(3)-equivariant graph neural networks. The resulting learning problem respects both the locality of the underlying electronic structure and the geometric transformation laws imposed by three-dimensional space.

The software is motivated by three specific design goals. First, it should be modular at the level of both scientific abstractions and code organization. Data ingestion, basis conversion, graph construction, irrep mapping, model definition, and training logic should be independently replaceable while still composing into a single workflow. Second, the framework should support supervision beyond raw matrix reconstruction. When Hamiltonian, overlap, and density matrices are predicted jointly, physically relevant observables such as energies and electron counts can be computed directly from the predicted operators, while atomic forces and, for periodic systems, stress tensors follow from differentiation with respect to atomic coordinates and cell vectors. Third, the framework should make architecture exploration an inherent capability rather than an afterthought. Choices such as irreducible feature content, message-passing depth, tensor-product design, head structure, resid-

ual connections, normalization, and observable losses must all be accessible through a coherent configuration and sweep interface. Related operator-learning efforts include the DeepH family of Hamiltonian-learning models for localized-basis DFT, including their original message-passing formulation, extensions to magnetic superstructures, an explicitly E(3)-equivariant formulation, hybrid-functional Hamiltonians, and conversion pathways from plane-wave data to localized-orbital Hamiltonians [15–20]. These works demonstrate the promise of Hamiltonian learning at scale and provide a close point of comparison for the operator-centric aims of **Mandala**, while our emphasis here is on a reusable software framework that unifies sparse operator prediction, observable guidance, multi-backend data handling, and broad architecture exploration.

These goals distinguish **Mandala** from earlier machine-learning workflows in two ways. Relative to purely observable-driven interatomic models, **Mandala** preserves an operator-level representation that remains useful for electronic-structure analysis. Relative to monolithic research scripts built around one specific dataset or one fixed neural architecture, it provides a reusable framework in which data formats, physical targets, and model variants can be exchanged without rewriting the entire code path. In this sense, **Mandala** is intended as a unified software environment for sparse operator learning, observable guidance, and architecture exploration.

The following sections describe the sparse matrix data model, the E(3)-equivariant learning formulation, the observable-aware loss construction, the force-training pathway, and the software abstractions that make these components composable. Numerical benchmarks on [TODO: list benchmark systems] and application-specific results for [TODO: list target applications] are reported in Section [TODO: numerical results section]. Comment: Let’s decide how to present the results...

2. Theoretical Background

2.1. From the full electron-nuclear Schrödinger equation to Kohn-Sham DFT

At the most fundamental nonrelativistic level, an atomistic system is described by the stationary many-body Schrödinger equation for all electrons and all nuclei,

$$\hat{H}(\mathbf{r}; \mathbf{R})\Psi(\mathbf{r}; \mathbf{R}) = E\Psi(\mathbf{r}; \mathbf{R}), \quad (1)$$

where $\Psi(\mathbf{r}; \mathbf{R})$ denotes the many-body wavefunction, $\mathbf{r}$ the collection of electronic coordinates, and $\mathbf{R}$ the collection of ionic coordinates. The semicolon

indicates that the dependence on ionic coordinates is parametric within the Born-Oppenheimer approximation. We adopt atomic units throughout, with $\hbar = m_e = e^2 = 1$, so that energies are measured in Hartree and lengths in Bohr radii.

Under the Born-Oppenheimer approximation [21], the electronic Hamiltonian is written as

$$\hat{H}(\mathbf{r}; \mathbf{R}) = \hat{T}_e(\mathbf{r}) + \hat{V}_{ee}(\mathbf{r}) + \hat{V}_{ei}(\mathbf{r}; \mathbf{R}) + E_{ii}(\mathbf{R}), \quad (2)$$

with

$$\hat{T}_e(\mathbf{r}) = -\frac{1}{2} \sum_{j=1}^{N_e} \nabla_j^2, \quad (3)$$

$$\hat{V}_{ee}(\mathbf{r}) = \frac{1}{2} \sum_{j=1}^{N_e} \sum_{k \neq j}^{N_e} \frac{1}{|\mathbf{r}_j - \mathbf{r}_k|}, \quad (4)$$

$$\hat{V}_{ei}(\mathbf{r}; \mathbf{R}) = -\sum_{j=1}^{N_e} \sum_{\alpha=1}^{N_i} \frac{Z_\alpha}{|\mathbf{r}_j - \mathbf{R}_\alpha|}, \quad (5)$$

and

$$E_{ii}(\mathbf{R}) = \frac{1}{2} \sum_{\alpha=1}^{N_i} \sum_{\beta \neq \alpha}^{N_i} \frac{Z_\alpha Z_\beta}{|\mathbf{R}_\alpha - \mathbf{R}_\beta|}. \quad (6)$$

The ion-ion term amounts to a constant energy shift for fixed ionic configuration, while the Hamiltonian and wavefunction depend only parametrically on $\mathbf{R}$. In the following, this parametric dependence is not always written explicitly.

The central remaining difficulty is therefore the interacting many-electron problem defined by $\hat{T}_e + \hat{V}_{ee} + \hat{V}_{ei}$ at fixed ionic geometry.

Density functional theory (DFT) replaces the many-electron wavefunction by the electron density as the central variable [1]. In the Kohn-Sham construction [2], the interacting electron problem is mapped onto a fictitious system of non-interacting electrons governed by

$$\left[-\frac{1}{2} \nabla^2 + v_S(\mathbf{r}) \right] \psi_j(\mathbf{r}) = \epsilon_j \psi_j(\mathbf{r}), \quad (7)$$

where $j = 1, \dots, N_e$ labels the Kohn-Sham orbitals. The corresponding electronic density is

$$n(\mathbf{r}) = \sum_{j=1}^{N_e} |\psi_j(\mathbf{r})|^2. \quad (8)$$

The Kohn-Sham potential is

$$v_S(\mathbf{r}) = \frac{\delta E_H[n]}{\delta n(\mathbf{r})} + \frac{\delta E_{XC}[n]}{\delta n(\mathbf{r})} + v_{ei}(\mathbf{r}), \quad (9)$$

where $E_H[n]$ is the Hartree energy, $E_{XC}[n]$ the exchange-correlation energy, and

$$v_{ei}(\mathbf{r}) = - \sum_{\alpha} \frac{Z_{\alpha}}{|\mathbf{r} - \mathbf{R}_{\alpha}|}. \quad (10)$$

Although the Kohn-Sham construction is formally exact, practical calculations rely on approximations to E_{XC} .

In this notation, the ground-state energy functional is written as

$$E[n] = T_S[n] + E_H[n] + E_{XC}[n] + \int d\mathbf{r} n(\mathbf{r}) v_{ei}(\mathbf{r}) + E_{ii}, \quad (11)$$

where $T_S[n]$ is the Kohn-Sham kinetic energy. This is the DFT-level starting point that **Mandala** connects to localized-orbital operator learning.

Mandala operates on localized-orbital matrix representations of this problem. Expanding the Kohn-Sham orbitals in a finite basis $\{\chi_{\mu}\}$,

$$\psi_j(\mathbf{r}) = \sum_{\mu} C_{\mu j} \chi_{\mu}(\mathbf{r}), \quad (12)$$

leads to the generalized eigenvalue problem

$$\sum_{\nu} H_{\mu\nu} C_{\nu j} = \epsilon_j \sum_{\nu} S_{\mu\nu} C_{\nu j}, \quad (13)$$

with

$$H_{\mu\nu} = \langle \chi_{\mu} | \hat{H}_{KS} | \chi_{\nu} \rangle, \quad S_{\mu\nu} = \langle \chi_{\mu} | \chi_{\nu} \rangle. \quad (14)$$

The one-particle density matrix is

$$D_{\mu\nu} = \sum_j f_j C_{\mu j} C_{\nu j}^*, \quad (15)$$

where f_j denotes the orbital occupation. In a nonorthogonal basis, the electron count is

$$N_e = \text{Tr}(DS), \quad (16)$$

while the energy observable used throughout **Mandala** is obtained from

$$E_{\text{op}} = \text{Tr}(DH). \quad (17)$$

In this work, **Mandala** uses the trace expressions above as operator-derived observables associated with the predicted matrices.

Localized basis sets also induce block sparsity. If orbital μ belongs to atom i and orbital ν belongs to atom j , then matrix entries can be organized into atom-pair blocks $X_{ij}^{\mathbf{L}}$, where $X \in \{H, S, D\}$ and $\mathbf{L} \in \mathbb{Z}^3$ indexes periodic lattice images. This sparse block structure is the operator domain learned by **Mandala**.

2.1.1. Hamiltonian gauge invariance and the overlap-shift correction

In a nonorthogonal basis, the generalized eigenproblem is invariant under a rigid shift of the Hamiltonian by a multiple of the overlap matrix. Let

$$HC = SC\varepsilon, \quad (18)$$

and define a shifted Hamiltonian

$$\tilde{H} = H - \alpha S. \quad (19)$$

Then

$$\tilde{H}C = (H - \alpha S)C = SC\varepsilon - \alpha SC = SC(\varepsilon - \alpha I). \quad (20)$$

Therefore the generalized eigenvectors are unchanged, while all eigenvalues are shifted by the same constant α . This proves that the choice of zero of energy can be represented, in matrix form, as the transformation $H \mapsto H - \alpha S$. For a fixed electron number and a geometry-independent shift α , one also finds

$$\text{Tr}(D\tilde{H}) = \text{Tr}(DH) - \alpha\text{Tr}(DS) = \text{Tr}(DH) - \alpha N_e, \quad (21)$$

so only the absolute energy reference changes.

Mandala exploits this gauge freedom through the overlap-shift correction μ_H , which is chosen to minimize the mean-squared discrepancy between a predicted Hamiltonian and a reference Hamiltonian after subtraction of a multiple of the overlap matrix:

$$\mu_H = \arg \min_{\mu} \|H^{\text{pred}} - H^{\text{ref}} - \mu S\|_F^2. \quad (22)$$

Taking the derivative with respect to μ and setting it to zero yields

$$\mu_H = \frac{\langle H^{\text{pred}} - H^{\text{ref}}, S \rangle_F}{\langle S, S \rangle_F}, \quad (23)$$

where $\langle A, B \rangle_F = \sum_{\mu\nu} A_{\mu\nu} B_{\mu\nu}$ is the Frobenius inner product, evaluated in **Mandala** blockwise over the aligned sparse representation. The corrected Hamiltonian $H^{\text{pred}} - \mu_H S$ therefore removes a pure gauge offset without altering the physically relevant eigenvectors.

2.2. $E(3)$ symmetry, equivariance, and irreducible representations

Atomistic systems in three-dimensional space are naturally acted on by the Euclidean group $E(3) = \mathbb{R}^3 \rtimes O(3)$, combining translations, rotations, and reflections. For a group element $g = (Q, \mathbf{t})$, with $Q \in O(3)$ and $\mathbf{t} \in \mathbb{R}^3$, atomic positions transform as

$$\mathbf{R}_i \mapsto g \cdot \mathbf{R}_i = Q \mathbf{R}_i + \mathbf{t}. \quad (24)$$

A model output $f(X)$ is invariant if $f(g \cdot X) = f(X)$, and equivariant if

$$f(g \cdot X) = \rho_{\text{out}}(g) f(X), \quad (25)$$

for a representation ρ_{out} appropriate to the predicted object. Scalar observables such as total energies are invariant. Vectors such as forces transform covariantly, and operator-valued quantities inherit structured transformation rules from the orbital basis in which they are expressed.

In equivariant neural networks, hidden features are not treated as generic channels. Instead, they are decomposed into irreducible representations (irreps) of $O(3)$, labeled by angular momentum ℓ and parity $p \in \{+1, -1\}$. A feature space therefore takes the form

$$\mathcal{H} = \bigoplus_{\ell, p} m_{\ell, p} D^{(\ell, p)}, \quad (26)$$

where $m_{\ell, p}$ is the multiplicity of irrep (ℓ, p) and $D^{(\ell, p)}$ is the corresponding representation matrix. Scalars correspond to $(\ell = 0, p = +1)$, ordinary vectors to $(\ell = 1, p = -1)$, and higher-rank objects are built from larger ℓ channels. This representation-theoretic bookkeeping is particularly natural for localized atomic orbitals, whose angular content is already organized by orbital angular momentum.

The key architectural consequence is that every learned operation must respect the decomposition into irreps. Node and edge features are updated through maps that commute with the group action, rather than by unrestricted dense linear layers. Geometric information enters through relative displacement vectors and their spherical-harmonic expansions, while translational invariance is enforced by expressing all geometry in relative coordinates. Reflections are tracked through the parity label p , while rotations act within each ℓ channel. In this way, **Mandala** preserves directional information throughout the network without sacrificing the required covariance laws [13, 14].

2.3. Tensor products, spherical harmonics, and the Wigner-Eckart viewpoint

The basic algebraic operation of an equivariant graph neural network is the tensor product of irreps. If two feature channels transform according to $D^{(\ell_1, p_1)}$ and $D^{(\ell_2, p_2)}$, then their tensor product decomposes as

$$D^{(\ell_1, p_1)} \otimes D^{(\ell_2, p_2)} = \bigoplus_{L=|\ell_1-\ell_2|}^{\ell_1+\ell_2} D^{(L, p_1 p_2)}. \quad (27)$$

This Clebsch-Gordan structure determines which output irreps may be produced by combining two input channels. In practical terms, it is the algebraic rule that governs message passing, nonlinear mixing of hidden features, and the final projection from latent features to matrix coefficients.

Edge geometry is injected through spherical harmonics $Y_{\ell m}(\hat{\mathbf{r}}_{ij})$, evaluated on normalized relative displacement directions. Because spherical harmonics transform irreducibly under rotations, they provide the canonical angular basis for equivariant edge messages. A typical equivariant interaction therefore combines learned node or edge features with spherical-harmonic features through tensor products, while radial functions modulate the interaction strength as a function of distance.

The same representation-theoretic structure appears in matrix elements of operators between angular-momentum-carrying orbitals. For a spherical tensor operator $T_q^{(k)}$, the Wigner-Eckart theorem states that

$$\langle \alpha \ell m | T_q^{(k)} | \alpha' \ell' m' \rangle = \langle \ell' m'; k q | \ell m \rangle \langle \alpha \ell || T^{(k)} || \alpha' \ell' \rangle, \quad (28)$$

where ℓ and ℓ' denote orbital angular momenta, m and m' their magnetic quantum numbers, $T_q^{(k)}$ is the q th component of a spherical tensor operator of

rank k , α and α' collect any additional quantum numbers needed to specify the states, and $\langle \ell' m'; kq | \ell m \rangle$ is a Clebsch-Gordan coefficient. The double-bar matrix element $\langle \alpha \ell || T^{(k)} || \alpha' \ell' \rangle$ is the corresponding reduced matrix element. that is, the dependence on magnetic quantum numbers is separated from the reduced matrix element. **Mandala** uses this viewpoint operationally: the angular coupling structure is fixed analytically by the irrep mapping, while the neural network learns the reduced coefficients associated with each orbital pair and chemical environment. This is precisely why the matrix-block prediction problem can be posed naturally in an irrep basis rather than directly in Cartesian matrix entries.

2.4. From equivariant local features to sparse operator prediction

Mandala represents each operator $X \in \{H, S, D\}$ as a sparse family of atom-pair blocks X_{ij}^L , where i and j index atoms and L labels periodic images. The block representation is converted into an irrep representation by a fixed change of basis,

$$\mathbf{x}_{ij}^L = Q_{a_i a_j} \text{vec}(X_{ij}^L), \quad (29)$$

where a_i and a_j denote the atomic species of atoms i and j , and $Q_{a_i a_j}$ is the species-pair-specific block-to-irrep transform. The inverse map reconstructs matrix blocks from predicted irrep vectors,

$$\text{vec}(X_{ij}^L) = Q_{a_i a_j}^{-1} \mathbf{x}_{ij}^L, \quad (30)$$

which, in the orthonormal convention used internally, is implemented as a transpose. In the code these maps are materialized by the **BlockIrrepMapper**. The same component is reused by the data pipeline, the model heads, and the evaluation routines.

Once the network predicts a consistent set of sparse operators, observable quantities are obtained directly from those operators rather than through a separate surrogate model. For sparse block matrices, the trace of a product is evaluated as

$$\text{Tr}(AB) = \sum_{(L,i,j)} \text{tr}(A_{ij}^L B_{ji}^{-L}), \quad (31)$$

which is exactly the sparse contraction implemented by **Mandala**. This gives direct access to electron counts, energy observables, and derivative observables such as forces and stress, all from the same predicted operator set. The

resulting workflow differs from direct scalar regression: internal consistency between matrices and observables is built into the model formulation rather than imposed only at the level of post hoc analysis.

3. Numerical Implementation

3.1. Software design goals, sparse data model, and periodic graph construction

Mandala is organized as a layered scientific software stack whose core workflow consists of parsing electronic-structure outputs, harmonizing basis conventions, building sparse operators and graphs, predicting irrep vectors, reconstructing operators, and evaluating observables. Figure 1 summarizes this end-to-end workflow, while Figure 2 highlights the data-loading and sample-construction path in more detail. The central data abstractions are **BlockMatrix**, **IrrepsBlockData**, and **Snapshot**. A **Snapshot** contains all matrices and metadata associated with one geometry, including positions, cell, optional forces, and optional stress targets. This keeps Hamiltonian, overlap, and density predictions tied to a single physically consistent basis and sparse indexing scheme.

The sparse operator representation stores blocks by ordered species pair and by explicit edge metadata $(\mathbf{L}, i, j)$, where $\mathbf{L} \in \mathbb{Z}^3$ is the lattice-image shift. Basis-conversion modules map backend-specific orbital orderings and sign conventions into the common internal representation expected by the equivariant stack. In the current framework, OpenMX, FHI-aims, and PySCF are treated as peer electronic-structure backends [22–25], with PySCF integration still under active development [TODO: finish PySCF training-data integration].

Graph construction mirrors the sparse operator indexing. Nodes correspond to atoms, while edges correspond to self-interactions and neighbor interactions inside a cutoff radius. For periodic systems, each edge carries an integer shift vector $\mathbf{L}$ and a relative displacement

$$\mathbf{r}_{ij}^{\mathbf{L}} = \mathbf{R}_j - \mathbf{R}_i + \sum_{\alpha=1}^3 L_{\alpha} \mathbf{a}_{\alpha}, \quad (32)$$

where $\{\mathbf{a}_{\alpha}\}$ are the cell vectors. These displacements are used to construct radial embeddings and spherical-harmonic features. The directed periodic

edge set is therefore

$$\mathcal{E} = \left\{ (i, j, \mathbf{L}) \mid \|\mathbf{r}_{ij}^{\mathbf{L}}\|_2 \leq r_c \right\}, \quad (33)$$

with a further partition into diagonal, shifted-self, and off-diagonal edges,

$$\mathcal{E}_{\text{diag}} = \{(i, j, \mathbf{L}) \in \mathcal{E} \mid i = j, \mathbf{L} = \mathbf{0}\}, \quad (34)$$

$$\mathcal{E}_{\text{shifted_self}} = \{(i, j, \mathbf{L}) \in \mathcal{E} \mid i = j, \mathbf{L} \neq \mathbf{0}\}, \quad (35)$$

$$\mathcal{E}_{\text{offdiag}} = \{(i, j, \mathbf{L}) \in \mathcal{E} \mid i \neq j\}. \quad (36)$$

To preserve exact agreement between graph edges and sparse matrix blocks, **Mandala** imposes a canonical ordering of edges by distance and lattice shift, with reverse-edge lookup tables used for sparse traces and symmetrization. This explicit reverse alignment is essential for both physical correctness and reproducibility under periodic boundary conditions.

The same snapshot-level representation preserves access to operator-derived postprocessing. In particular, the predicted Hamiltonian may be reused for band-structure analysis within the same sparse-operator workflow [TODO: specify workflow/API entry point].

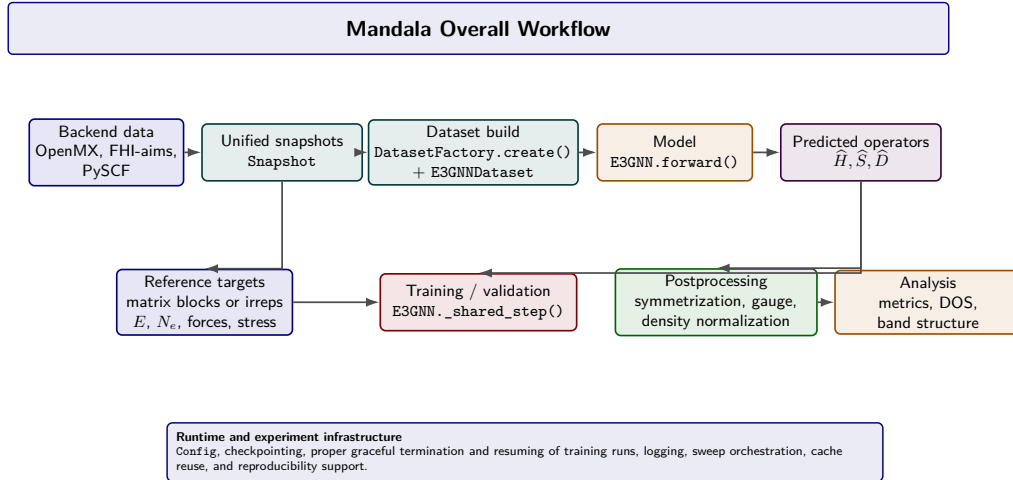


Figure 1: Overall **Mandala** workflow. Backend data are converted into unified **Snapshot** objects, transformed into graph-aligned datasets, processed by the equivariant model, and used both for training and for postprocessing and analysis.

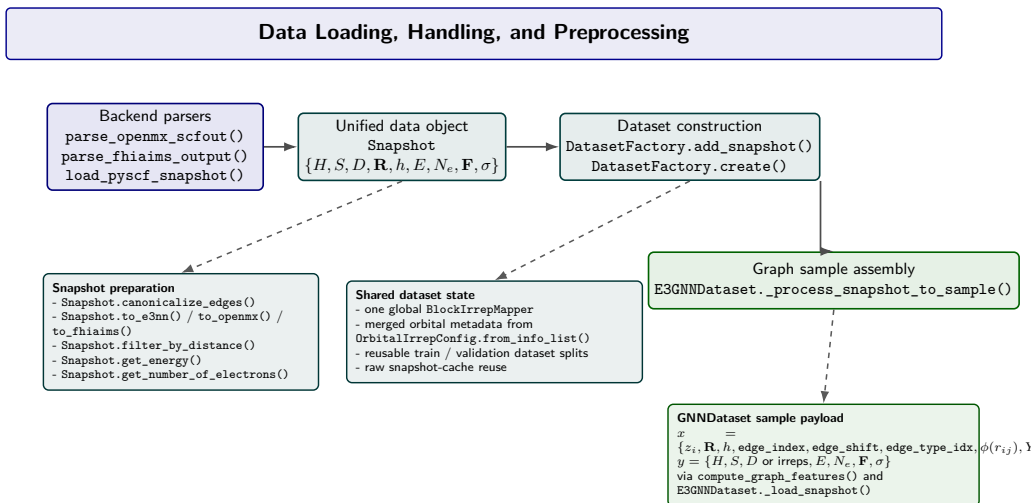


Figure 2: Data loading, handling, and preprocessing. The figure emphasizes the transition from backend-specific parsers to unified **Snapshot** objects, **DatasetFactory**-driven dataset construction, and **E3GNNDataset**-level graph sample assembly.

3.2. $E(3)$ -equivariant message passing, readout heads, and architecture variants

The network is composed of node encoders, edge encoders, a stack of equivariant message-passing blocks, and matrix-specific readout heads. Node encoders embed chemical species into scalar channels. Edge encoders combine edge-type embeddings, radial distance embeddings, and spherical harmonics into equivariant edge features. Figure 3 summarizes the computational structure of the model stack, and Figure 6 summarizes the configuration-driven search space exposed by the software. If z_i denotes the atomic-species index of atom i , the initial node representation is

$$\mathbf{h}_i^{(0)} = \text{Emb}_{\text{atom}}(z_i) \in d_0 \times 0e, \quad (37)$$

while one available edge-encoder variant first forms an edge-type embedding $\mathbf{u}_{ij} = \text{Emb}_{\text{edge}}(t_{ij})$ and a geometric feature $\mathbf{g}_{ij} = \varphi(r_{ij}) \oplus Y(\hat{\mathbf{r}}_{ij})$, and then applies

$$\mathbf{e}_{ij}^{(0)} = \text{TP}_{\text{enc}}(\mathbf{u}_{ij}, \mathbf{g}_{ij}). \quad (38)$$

If $\mathbf{h}_i^{(t)}$ denotes node features and $\mathbf{e}_{ij}^{(t)}$ denotes edge features at layer t , a generic **Mandala** message-passing step may be written as

$$\mathbf{m}_{ij}^{(t)} = \Phi_{\text{edge}}^{(t)} \left(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}, Y(\hat{\mathbf{r}}_{ij}), \varphi(r_{ij}) \right), \quad (39)$$

$$\mathbf{h}_i^{(t+1)} = \Phi_{\text{node}}^{(t)} \left(\mathbf{h}_i^{(t)}, \text{Agg}_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(t)} \right), \quad (40)$$

Here, $\mathbf{h}_i^{(t)}$ denotes the irrep-valued latent feature carried by node i at message-passing layer t , $\mathbf{e}_{ij}^{(t)}$ denotes the corresponding edge feature for the directed edge $i \rightarrow j$, $\mathbf{m}_{ij}^{(t)}$ is the edge message constructed at that layer, $\hat{\mathbf{r}}_{ij}$ is the unit vector along the relative displacement between atoms i and j , $Y(\hat{\mathbf{r}}_{ij})$ denotes the spherical-harmonic embedding of that direction, and $\varphi(r_{ij})$ denotes a radial distance embedding. The neighbor set $\mathcal{N}(i)$ contains the atoms connected to node i within the graph cutoff. where $\Phi_{\text{edge}}^{(t)}$ and $\Phi_{\text{node}}^{(t)}$ are equivariant maps assembled from tensor products, equivariant linear layers, radial MLPs, parity-aware nonlinearities, and optional residual connections, while Agg denotes a configurable aggregation rule such as summation or attention.

In the concrete implementation, these updates are built from equivariant convolutions with radial weighting. Writing the concatenated local feature as

$$\mathbf{f}_{ij}^{(t)} = \mathbf{h}_i^{(t)} \oplus \mathbf{h}_j^{(t)} \oplus \mathbf{e}_{ij}^{(t)}, \quad (41)$$

the tensor-product stage produces

$$\mathbf{z}_{ij}^{(t)} = \text{TP} \left(\mathbf{f}_{ij}^{(t)}, Y(\hat{\mathbf{r}}_{ij}) \right), \quad (42)$$

followed by an optional equivariant nonlinearity

$$\tilde{\mathbf{z}}_{ij}^{(t)} = \text{Act}_{\text{eq}} \left(\mathbf{z}_{ij}^{(t)} \right), \quad (43)$$

and a radial MLP that returns one scalar coefficient per output irrep block,

$$\mathbf{w}_{ij}^{(t)} = \text{MLP}_{\text{rad}}(\varphi(r_{ij})). \quad (44)$$

If $\tilde{\mathbf{z}}_{ij}^{(t,a)}$ denotes the a th output irrep slice, the weighted equivariant convolution is

$$\mathbf{q}_{ij}^{(t,a)} = w_{ij}^{(t,a)} \tilde{\mathbf{z}}_{ij}^{(t,a)}. \quad (45)$$

The node update then aggregates edge messages as

$$\overline{\mathbf{m}}_i^{(t)} = \sum_{j \in \mathcal{N}(i)} \mathbf{q}_{ji}^{(t)}, \quad (46)$$

while edge and node updates both admit optional self-connections and residual paths. In the sequential message block used by the code, node features are first updated from the current edges and only then are edge features updated using the new node state.

The readout stage maps final latent features to the irrep content required by each matrix target. For a target operator X , the head predicts irrep vectors

$$\widehat{\mathbf{x}}_{ij}^{\mathbf{L},X} = \Psi_X\left(\mathbf{e}_{ij}^{(L)}, \mathbf{h}_i^{(L)}, \mathbf{h}_j^{(L)}, \varphi(r_{ij})\right), \quad (47)$$

followed by inverse irrep-to-block conversion

$$\widehat{X}_{ij}^{\mathbf{L}} = Q_{a_i a_j}^\top \widehat{\mathbf{x}}_{ij}^{\mathbf{L},X}. \quad (48)$$

For richer readout variants, the edge used by the head may first be transformed through a tensor-square map,

$$\mathbf{t}_{ij}^{(L)} = \text{TS}\left(\mathbf{e}_{ij}^{(L)}\right), \quad (49)$$

with the actual head input defined by

$$\mathbf{u}_{ij}^{(L)} = \begin{cases} \mathbf{t}_{ij}^{(L)}, & \text{tensor-square head enabled,} \\ \mathbf{e}_{ij}^{(L)}, & \text{otherwise.} \end{cases} \quad (50)$$

The head can then branch by edge class,

$$\widehat{\mathbf{x}}_{ij}^{\mathbf{L},X} = \begin{cases} \Psi_{\text{diag}}^X(\mathbf{u}_{ij}^{(L)}), & (i, j, \mathbf{L}) \in \mathcal{E}_{\text{diag}}, \\ \Psi_{\text{shifted_self}}^X(\mathbf{u}_{ij}^{(L)}), & (i, j, \mathbf{L}) \in \mathcal{E}_{\text{shifted_self}}, \\ \Psi_{\text{offdiag}}^X(\mathbf{u}_{ij}^{(L)}), & (i, j, \mathbf{L}) \in \mathcal{E}_{\text{offdiag}}. \end{cases} \quad (51)$$

When the onsite node-embedding override is enabled, zero-shift self-edges use

$$\mathbf{u}_{ii}^0 \leftarrow \mathbf{h}_i^{(L)} \quad (52)$$

instead of the corresponding edge embedding. The most elaborate readout variants further split scalar and non-scalar output channels or factorize each

predicted block into a normalized equivariant direction and a positive scalar magnitude. In the latter case one predicts

$$\hat{\mathbf{x}}_{e,\text{norm}}^X = \Psi_{\text{dir}}^X(\mathbf{u}_e), \quad \hat{m}_e^X = \exp(\text{MLP}_{\text{mag}}^X(L_{\text{mag}}^X(\mathbf{t}_e) \oplus \varphi(r_e))), \quad (53)$$

and reconstructs the actual block prediction as

$$\hat{X}_e^{\text{actual}} = \hat{m}_e^X \hat{X}_e^{\text{norm}}. \quad (54)$$

Because heads share the same equivariant latent representation, **Mandala** can predict Hamiltonian, overlap, and density matrices jointly with only modest additional readout cost.

An important feature of the framework is that the architecture is not fixed to one canonical model. **Mandala** exposes a broad family of variants within the same software interface. Search dimensions include the hidden irrep content, the angular cutoff ℓ_{max} , the number of message-passing layers, the radial basis size, the depth of the neck and head MLPs, and the choice of tensor product implementation. The message-passing blocks support alternative edge updates (tensor-product, concatenation-based, and replacement-style variants), alternative node updates (sum, concatenation, and replacement variants), alternative aggregation mechanisms (including attention-style aggregation), pre- and post-linear transformations, optional self-connections, and optional node and edge residual paths.

The encoder and head stages are likewise configurable. Edge encoders support various variants, together with optional tensor-square feature expansion of spherical-harmonics. Readout heads support direct equivariant projections, deeper equivariant MLP heads, optional tensor-square preconditioning of hidden features, scalar-specific auxiliary branches, optional log-scale or magnitude factorization for output amplitudes, optional use of node embeddings for self-edges, and separate treatment of diagonal, shifted-self, and off-diagonal blocks. Because all of these variants share the same block-sparse data model and irrep mapping layer, architecture search can be performed without changing the surrounding training and evaluation workflow.

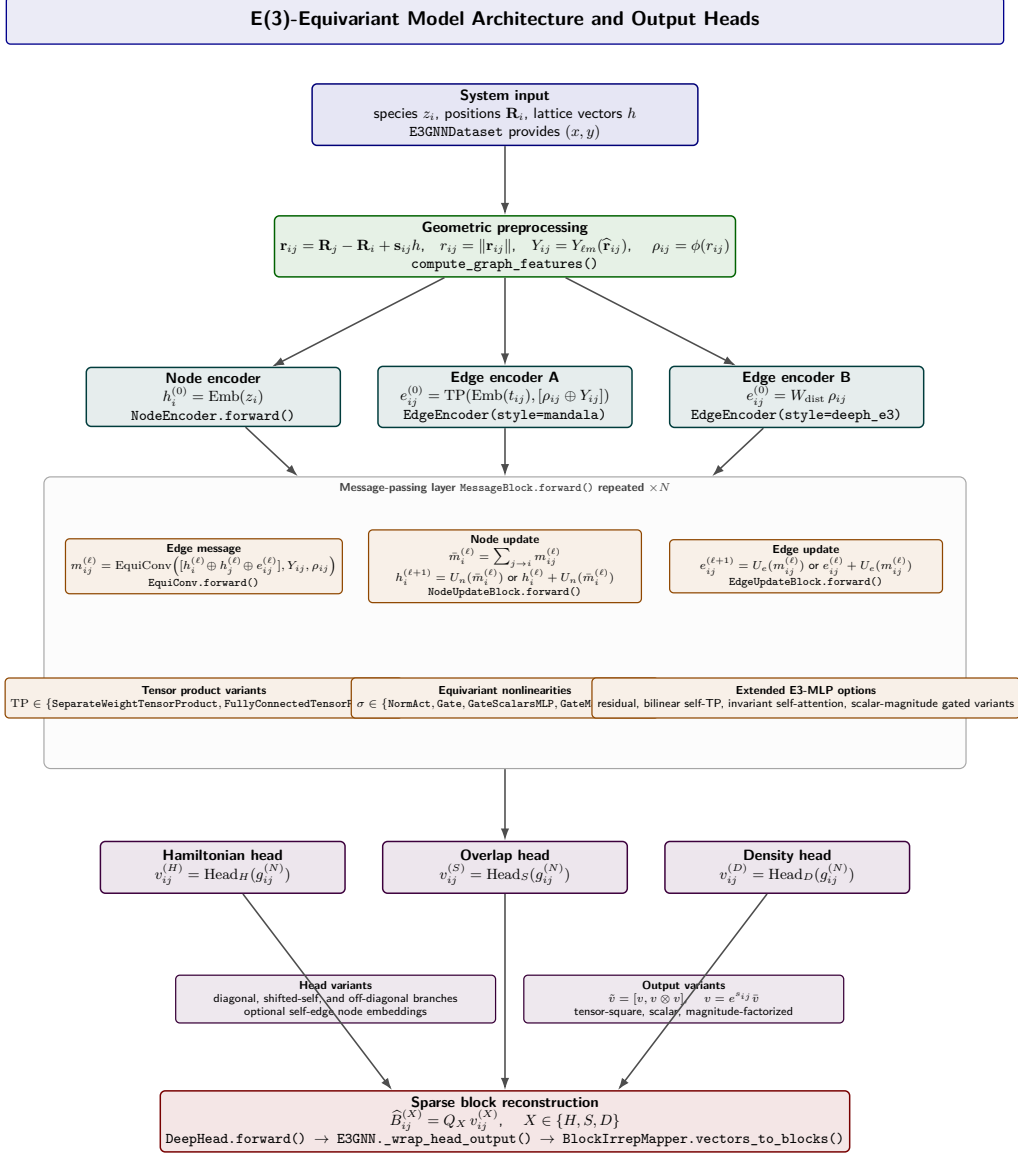


Figure 3: Computational structure of the E(3)-equivariant model. System inputs are encoded into node and edge features, propagated through repeated equivariant message-passing blocks, and mapped by operator-specific heads back to sparse matrix blocks.

3.3. Sparse operator algebra, symmetrization, gauge correction, and density normalization

Sparse operator manipulations are implemented directly on aligned block data. Given two sparse operators A and B with reverse-edge alignment already established, **Mandala** evaluates their trace contraction as

$$\text{Tr}(AB) = \sum_{(\mathbf{L}, i, j)} \text{tr}(A_{ij}^{\mathbf{L}} B_{ji}^{-\mathbf{L}}), \quad (55)$$

without assembling dense global matrices. This operator algebra is reused for energy and electron-count evaluation, loss construction, and diagnostic analysis. For efficient vectorized evaluation, the implementation precomputes a reverse-edge alignment permutation. For each pair key k with reverse key k^{rev} , the permutation P_k is defined by

$$P_k(n) = m \iff (\mathbf{L}_{k,n}, i_{k,n}, j_{k,n}) = (-\mathbf{L}_{k^{\text{rev}},m}, j_{k^{\text{rev}},m}, i_{k^{\text{rev}},m}), \quad (56)$$

so that the aligned trace may be written as

$$\text{Tr}(AB) = \sum_k \sum_n \text{tr}(A_{k,n} B_{k^{\text{rev}}, P_k(n)}). \quad (57)$$

For matrix targets expected to be symmetric or Hermitian in the chosen basis, **Mandala** supports explicit sparse symmetrization after block reconstruction. In the real-valued convention used by the implementation, this takes the form

$$X_{ij}^{\mathbf{L}} \leftarrow \frac{1}{2} (X_{ij}^{\mathbf{L}} + (X_{ji}^{-\mathbf{L}})^{\top}). \quad (58)$$

This operation is carried out using precomputed reverse-edge permutations, so that symmetry is enforced consistently across periodic images and self-edges.

Hamiltonian gauge freedom is handled operationally through the overlap-shift correction described in Section 2.1.1. In practical evaluation and analysis workflows, **Mandala** computes

$$\tilde{H}^{\text{pred}} = H^{\text{pred}} - \mu_H S, \quad \mu_H = \frac{\langle H^{\text{pred}} - H^{\text{ref}}, S \rangle_F}{\langle S, S \rangle_F}, \quad (59)$$

so that purely gauge-like Hamiltonian offsets are removed before reporting gauge-corrected operator errors.

The framework also supports density normalization to the correct number of electrons. If a predicted density $\hat{D}$ and overlap $\hat{S}$ produce $\hat{N}_e = \text{Tr}(\hat{D}\hat{S})$, then a normalized density is obtained by the scalar rescaling

$$\hat{D}_{\text{norm}} = \beta \hat{D}, \quad \beta = \frac{N_e^{\text{target}}}{\text{Tr}(\hat{D}\hat{S})}. \quad (60)$$

This provides a simple and effective way to enforce electron-count consistency.

Together, sparse trace evaluation, symmetry restoration, gauge correction, and density normalization define a compact postprocessing layer. It turns raw network outputs into physically interpretable operators ready for observable evaluation and Hamiltonian-based analyses such as band-structure calculation.

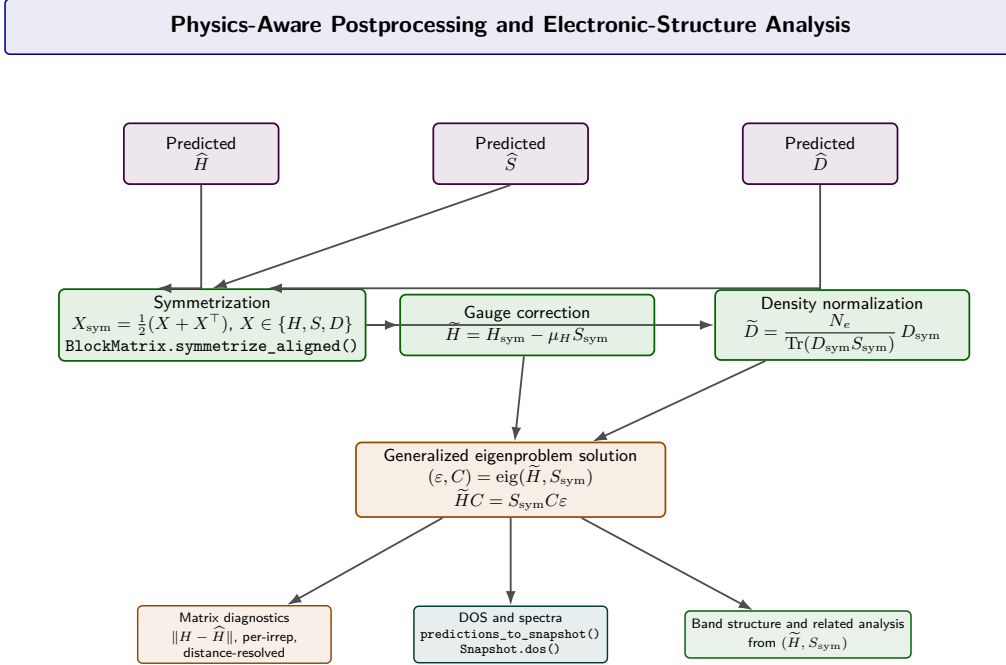


Figure 4: Physics-aware postprocessing and electronic-structure analysis from predicted sparse operators. The workflow makes the roles of $\hat{H}$, $\hat{S}$, and $\hat{D}$ explicit and shows how symmetrization, gauge correction, density normalization, and generalized-eigenproblem solution connect to diagnostics and spectral analysis.

3.4. Observable guidance with energies, electron counts, forces, and stress

Mandala supports pure matrix learning, mixed matrix-and-observable learning, and fully observable-guided training from one common operator-prediction pipeline. Figure 5 summarizes the corresponding training flow. For a chosen set of matrix targets $\mathcal{M} \subseteq \{H, S, D\}$, the general training objective is

$$\mathcal{L} = \sum_{X \in \mathcal{M}} \lambda_X \mathcal{L}_X^{\text{matrix}} + \lambda_E \mathcal{L}^E + \lambda_N \mathcal{L}^N + \lambda_F \mathcal{L}^F + \lambda_\sigma \mathcal{L}^\sigma. \quad (61)$$

The matrix terms compare predicted and reference sparse operators, typically by a weighted combination of ℓ_1 and ℓ_2 errors,

$$\mathcal{L}_X^{\text{matrix}} = \eta \text{MAE}(\hat{X}, X) + (1 - \eta) \text{MSE}(\hat{X}, X), \quad (62)$$

with η controlled by configuration.

Observable guidance is defined on quantities derived from the predicted operators. In the operator-centered workflow used throughout **Mandala**, the energy observable used in training and the number of electrons are

$$\hat{E} = \text{Tr}(\hat{D}\hat{H}), \quad \hat{N}_e = \text{Tr}(\hat{D}\hat{S}), \quad (63)$$

which gives

$$\mathcal{L}^E = \text{MSE}(\hat{E}, E^{\text{ref}}), \quad \mathcal{L}^N = \text{MSE}(\hat{N}_e, N_e^{\text{ref}}). \quad (64)$$

In the sparse aligned implementation, these quantities are evaluated as

$$\hat{E} = \sum_k \sum_n \text{tr}(\hat{H}_{k,n} \hat{D}_{k^{\text{rev}}, P_k(n)}), \quad (65)$$

$$\hat{N}_e = \sum_k \sum_n \text{tr}(\hat{D}_{k,n} \hat{S}_{k^{\text{rev}}, P_k(n)}). \quad (66)$$

The framework also supports mixed observable guidance paths in which one of the operators entering the trace is replaced by the reference operator, for example the diagnostic contractions

$$\hat{E}_{H|D^{\text{ref}}} = \text{Tr}(\hat{H} D^{\text{ref}}), \quad \hat{E}_{D|H^{\text{ref}}} = \text{Tr}(H^{\text{ref}} \hat{D}), \quad (67)$$

and

$$\hat{N}_{D|S^{\text{ref}}} = \text{Tr}(\hat{D} S^{\text{ref}}), \quad \hat{N}_{S|D^{\text{ref}}} = \text{Tr}(D^{\text{ref}} \hat{S}), \quad (68)$$

enabling diagnostic and training modes that separate errors due to different predicted operators.

Forces are obtained by automatic differentiation of the predicted scalar energy with respect to atomic coordinates,

$$\hat{\mathbf{F}}_I = -\frac{\partial \hat{E}}{\partial \mathbf{R}_I}, \quad (69)$$

and the corresponding force loss is

$$\mathcal{L}^F = \frac{1}{3N_{\text{atoms}}} \sum_I \left\| \hat{\mathbf{F}}_I - \mathbf{F}_I^{\text{ref}} \right\|_2^2. \quad (70)$$

No separate force head is required: forces are constrained through the same predicted operators that define the energy.

For periodic systems, stress guidance is obtained by differentiating the predicted energy with respect to the cell matrix h . If $\Omega = \det(h)$ is the cell volume, **Mandala** uses

$$\hat{\sigma} = \frac{1}{\Omega} h^\top \frac{\partial \hat{E}}{\partial h}, \quad (71)$$

with optional symmetrization

$$\hat{\sigma} \leftarrow \frac{1}{2} (\hat{\sigma} + \hat{\sigma}^\top), \quad (72)$$

and defines the stress loss as

$$\mathcal{L}^\sigma = \frac{1}{9} \left\| \hat{\sigma} - \sigma^{\text{ref}} \right\|_F^2. \quad (73)$$

This completes the observable-guidance picture. A single operator-valued model can be trained under simultaneous constraints from matrix entries, energies, electron counts, forces, and stress tensors.

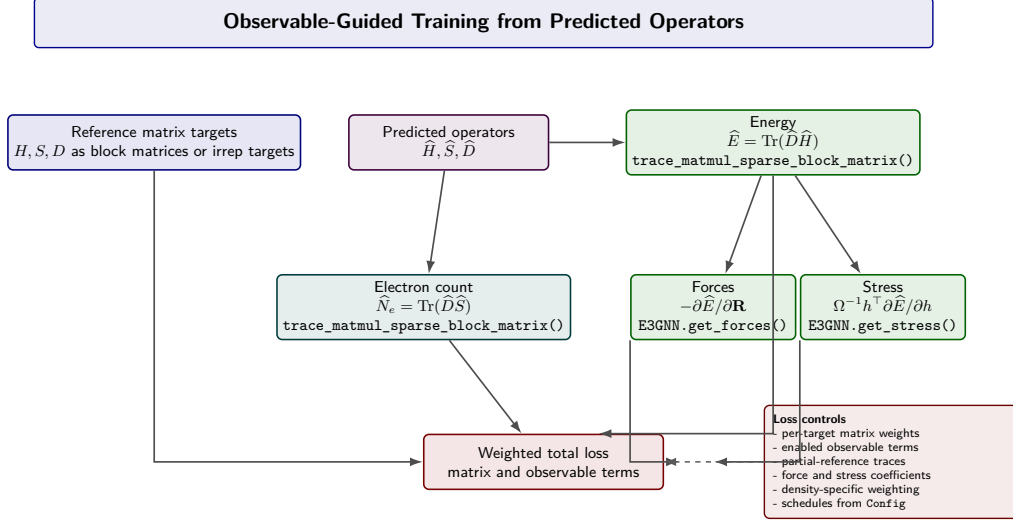


Figure 5: Observable-guided training from predicted operators. The same predicted operator set supports direct matrix supervision together with energy, electron-count, force, and stress guidance.

3.5. Configuration-driven architecture search

Because the same sparse operator workflow supports many architectural choices, **Mandala** treats model search as a primary scientific use case rather than as an auxiliary engineering convenience. All major representation, encoder, message-passing, readout, and loss-weighting choices are exposed as structured configuration parameters and can therefore be swept systematically.

Search dimensions include hidden irreps, angular cutoff, radial basis size, message-passing depth, encoder style, tensor-product type, normalization and nonlinearity families, node and edge update variants, residual pathways, neck and head depth, separate treatment of shifted-self blocks, tensor-square readout options, scalar-specific branches, observable-loss coefficients, force and stress weights, and the subset of matrix targets predicted jointly. This wide configuration surface is one of the main scientific strengths of the framework: it enables controlled comparisons between architecture families without forcing changes to the data model, parser layer, or observable-evaluation stack. The importance of systematic hyperparameter exploration in machine-learned electronic-structure models has also been emphasized in related work on training-free optimization strategies [26].

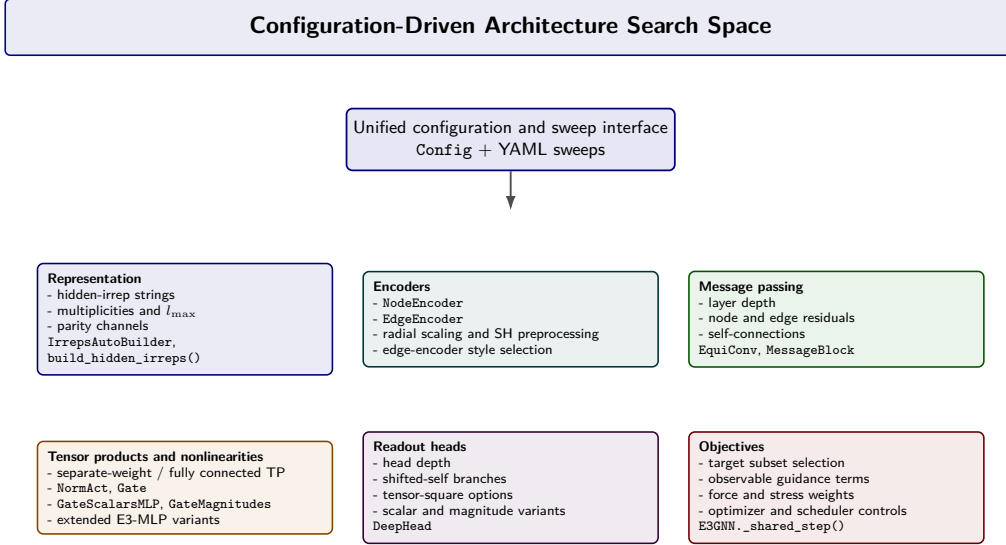


Figure 6: Configuration-driven architecture search space exposed by **Mandala**. Representation choices, encoders, message-passing layers, tensor-product variants, readout heads, and objective terms are controlled within one common configuration interface.

3.6. Verification, reproducibility, and extensibility

Mandala includes tests and validation routines covering data parsing, basis harmonization, sparse operator alignment, block-to-irrep mapping, graph construction, equivariant tensor-product paths, observable evaluation, and end-to-end training workflows. These checks are especially important in a code base where subtle mistakes in basis ordering, reverse-edge alignment, or irrep bookkeeping can produce apparently plausible but physically inconsistent results.

Reproducibility follows from configuration-driven runs, explicit snapshot serialization, deterministic sparse edge ordering, and checkpointed training. Extensibility follows from the layered structure described above: new electronic structure backends, new observable definitions, new equivariant blocks, and new training objectives can be incorporated without changing the core operator data model. This design is what allows **Mandala** to function both as a reusable scientific software package and as a framework for rapid methodological experimentation.

4. Conclusion

Mandala is a scientific software framework for learning sparse electronic-structure operators with $E(3)$ -equivariant graph neural networks. Its defining features are the explicit sparse block representation of Hamiltonian, overlap, and density matrices; the modular software layers that connect parsing, basis harmonization, irrep mapping, graph construction, model definition, and training; and the ability to supervise not only matrices themselves but also observables derived from them.

The framework is designed to connect two communities that are often separated in practice. From the electronic-structure side, it retains access to operator-level information and basis-aware quantities. From the atomistic machine-learning side, it inherits the flexibility of modern equivariant message passing, multitask learning, and automated hyperparameter search. By combining these perspectives, **Mandala** supports workflows in which one can train on matrix fidelity, energy-observable fidelity, electron-count fidelity, force fidelity, and stress fidelity within one coherent implementation.

Quantitative benchmarks on [TODO: list systems], scientific case studies for [TODO: list applications], and comparative assessments against [TODO: list baseline methods] are reported in Section [TODO: numerical results section]. **Mandala** is therefore positioned not merely as a prototype model, but as a modular platform for operator-level machine learning in electronic-structure simulation. In this sense, **Mandala** is aimed at simulation settings that go beyond conventional machine-learning interatomic potentials: it combines access to atomistic forces with access to the predicted electronic structure itself, including Hamiltonian-derived quantities such as band structure and related observables. This makes the framework particularly relevant for problems in which explicitly electronic effects, such as charge transfer or electronically driven changes in bonding, are part of the physics that must be modeled rather than treated only indirectly.

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References

- [1] P. Hohenberg, W. Kohn, Inhomogeneous electron gas, *Phys. Rev.* 136 (1964) B864–B871. doi:10.1103/PhysRev.136.B864.
- [2] W. Kohn, L. J. Sham, Self-Consistent Equations Including Exchange and Correlation Effects, *Phys. Rev.* 140 (1965) A1133–A1138. doi:10.1103/PhysRev.140.A1133.
- [3] L. Fiedler, K. Shah, M. Bussmann, A. Cangi, Deep dive into machine learning density functional theory for materials science and chemistry, *Phys. Rev. Mater.* 6 (2022) 040301. doi:10.1103/PhysRevMaterials.6.040301.
URL <https://link.aps.org/doi/10.1103/PhysRevMaterials.6.040301>
- [4] J. A. Ellis, L. Fiedler, G. A. Popoola, N. A. Modine, J. A. Stephens, A. P. Thompson, A. Cangi, S. Rajamanickam, Accelerating finite-temperature Kohn-Sham density functional theory with deep neural networks, *Phys. Rev. B* 104 (2021) 035120. doi:10.1103/PhysRevB.104.035120.
URL <https://link.aps.org/doi/10.1103/PhysRevB.104.035120>
- [5] L. Fiedler, N. A. Modine, K. D. Miller, A. Cangi, Machine learning the electronic structure of matter across temperatures, *Phys. Rev. B* 108 (2023) 125146. doi:10.1103/PhysRevB.108.125146.
URL <https://link.aps.org/doi/10.1103/PhysRevB.108.125146>
- [6] L. Fiedler, N. A. Modine, S. Schmerler, D. J. Vogel, G. A. Popoola, A. P. Thompson, S. Rajamanickam, A. Cangi, Predicting electronic structures at any length scale with machine learning, *npj Comput Mater* 9 (1) (2023) 1–10. doi:10.1038/s41524-023-01070-z.
URL <https://www.nature.com/articles/s41524-023-01070-z>
- [7] L. Fiedler, Z. A. Moldabekov, X. Shao, K. Jiang, T. Dornheim, M. Pavanello, A. Cangi, Accelerating equilibration in first-principles molecular dynamics with orbital-free density functional theory, *Phys. Rev. Res.* 4 (2022) 043033. doi:10.1103/PhysRevResearch.4.043033.
URL <https://link.aps.org/doi/10.1103/PhysRevResearch.4.043033>

- [8] S. A. Ghasemi, T. D. Kühne, Artificial neural networks for the kinetic energy functional of non-interacting fermions, *J. Chem. Phys.* 154 (7) (2021) 074107. doi:10.1063/5.0037319.
URL <https://doi.org/10.1063/5.0037319>
- [9] A. Chandrasekaran, D. Kamal, R. Batra, C. Kim, L. Chen, R. Ramprasad, Solving the electronic structure problem with machine learning, *npj computational materials* 5 (1) (2019) 22. doi:10.1038/s41524-019-0162-7.
- [10] M. Tsubaki, T. Mizoguchi, Quantum Deep Field: Data-Driven Wave Function, Electron Density Generation, and Atomization Energy Prediction and Extrapolation with Machine Learning, *Phys. Rev. Lett.* 125 (2020) 206401. doi:10.1103/PhysRevLett.125.206401.
URL <https://link.aps.org/doi/10.1103/PhysRevLett.125.206401>
- [11] X. Shao, L. Paetow, M. E. Tuckerman, M. Pavanello, Machine learning electronic structure methods based on the one-electron reduced density matrix, *Nature Communications* 14 (1) (Oct. 2023). doi:10.1038/s41467-023-41953-9.
URL <http://dx.doi.org/10.1038/s41467-023-41953-9>
- [12] F. Brockherde, L. Vogt, L. Li, M. E. Tuckerman, K. Burke, K.-R. Müller, Bypassing the Kohn-Sham equations with machine learning, *Nat. Commun.* 8 (1) (2017) 872. doi:10.1038/s41467-017-00839-3.
- [13] E. Kocer, T. W. Ko, J. Behler, Neural network potentials: A concise overview of methods, *Annual Review of Physical Chemistry* 73 (Volume 73, 2022) (2022) 163–186. doi:<https://doi.org/10.1146/annurev-physchem-082720-034254>.
URL <https://www.annualreviews.org/content/journals/10.1146/annurev-physchem-082720-034254>
- [14] J. A. Rackers, L. Tecot, M. Geiger, T. E. Smidt, A recipe for cracking the quantum scaling limit with machine learned electron densities, *Machine Learning: Science and Technology* 4 (1) (2023) 015027. doi:10.1088/2632-2153/acb314.
URL <https://dx.doi.org/10.1088/2632-2153/acb314>

- [15] H. Li, Z. Wang, N. Zou, M. Ye, R. Xu, X. Gong, W. Duan, Y. Xu, Deep-learning density functional theory hamiltonian for efficient ab initio electronic-structure calculation, *Nat. Comput. Sci.* 2 (6) (2022) 367–377. doi:10.1038/s43588-022-00265-6.
URL <https://doi.org/10.1038/s43588-022-00265-6>
- [16] H. Li, Z. Tang, X. Gong, N. Zou, W. Duan, Y. Xu, Deep-learning electronic-structure calculation of magnetic superstructures, *Nat. Comput. Sci.* 3 (4) (2023) 321–327. doi:10.1038/s43588-023-00424-3.
URL <https://doi.org/10.1038/s43588-023-00424-3>
- [17] X. Gong, H. Li, N. Zou, R. Xu, W. Duan, Y. Xu, General framework for e(3)-equivariant neural network representation of density functional theory hamiltonian, *Nat. Commun.* 14 (1) (2023) 2848. doi:10.1038/s41467-023-38468-8.
URL <https://doi.org/10.1038/s41467-023-38468-8>
- [18] Z. Tang, H. Li, P. Lin, X. Gong, G. Jin, L. He, H. Jiang, X. Ren, W. Duan, Y. Xu, A deep equivariant neural network approach for efficient hybrid density functional calculations, *Nat. Commun.* 15 (1) (2024) 8815. doi:10.1038/s41467-024-53028-4.
URL <https://doi.org/10.1038/s41467-024-53028-4>
- [19] X. Gong, S. G. Louie, W. Duan, Y. Xu, Generalizing deep learning electronic structure calculation to the plane-wave basis, *Nat. Comput. Sci.* 4 (10) (2024) 752–760. doi:10.1038/s43588-024-00701-9.
URL <https://doi.org/10.1038/s43588-024-00701-9>
- [20] Z. Tang, H. Chen, Y. Li, Y. Qian, Y. Wang, W. Fu, J. Li, C. Si, W. Duan, J. Chen, Y. Xu, Deep-learning electronic structure calculations, *Nat. Comput. Sci.* 5 (12) (2025) 1133–1146. doi:10.1038/s43588-025-00932-4.
URL <https://doi.org/10.1038/s43588-025-00932-4>
- [21] M. Born, R. Oppenheimer, Zur Quantentheorie der Molekeln, *Ann. Phys.* 389 (20) (1927) 457–484. doi:10.1002/andp.19273892002.
- [22] T. Ozaki, Variationally optimized atomic orbitals for large-scale electronic structures, *Phys. Rev. B* 67 (15) (2003) 155108. doi:10.1103/

PhysRevB.67.155108.

URL <https://doi.org/10.1103/PhysRevB.67.155108>

- [23] T. Ozaki, H. Kino, Numerical atomic basis orbitals from h to kr, Phys. Rev. B 69 (19) (2004) 195113. doi:10.1103/PhysRevB.69.195113.
URL <https://doi.org/10.1103/PhysRevB.69.195113>
- [24] V. Blum, R. Gehrke, F. Hanke, P. Havu, V. Havu, X. Ren, K. Reuter, M. Scheffler, Ab initio molecular simulations with numeric atom-centered orbitals, Comput. Phys. Commun. 180 (11) (2009) 2175–2196. doi:10.1016/j.cpc.2009.06.022.
URL <https://doi.org/10.1016/j.cpc.2009.06.022>
- [25] Q. Sun, X. Zhang, S. Banerjee, P. Bao, M. Barbry, N. S. Blunt, N. A. Bogdanov, G. H. Booth, J. Chen, Z.-H. Cui, J. J. Eriksen, Y. Gao, S. Guo, J. Hermann, M. R. Hermes, K. Koh, P. Koval, S. Lehtola, Z. Li, J. Liu, N. Mardirossian, J. D. McClain, M. Motta, B. Mussard, H. Q. Pham, A. Pulkin, W. Purwanto, P. J. Robinson, E. Ronca, E. R. Sayfutyarova, M. Scheurer, H. F. Schurkus, J. E. T. Smith, C. Sun, S.-N. Sun, S. Upadhyay, L. K. Wagner, X. Wang, A. White, J. D. Whitfield, M. J. Williamson, S. Wouters, J. Yang, J. M. Yu, T. Zhu, T. C. Berkelbach, S. Sharma, A. Y. Sokolov, G. K.-L. Chan, Recent developments in the PySCF program package, J. Chem. Phys. 153 (2) (2020) 024109. doi:10.1063/5.0006074.
URL <https://doi.org/10.1063/5.0006074>
- [26] L. Fiedler, N. Hoffmann, P. Mohammed, G. A. Popoola, T. Yovell, V. Oles, J. A. Ellis, S. Rajamanickam, A. Cangi, Training-free hyperparameter optimization of neural networks for electronic structures in matter, Mach. Learn.: Sci. Technol. 3 (4) (2022) 045008. doi:10.1088/2632-2153/ac9956.